VIT
Vellore Institute of Technology
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CHENNAI

B.TECH CSE (AI & ML) · BCSE497J – PROJECT 1

Macrograph AI

Multi-agent intelligence for the Indian economy, where every number traces back to an official source

10 sector agents | A2A + FastMCP | Causal knowledge graph

PRESENTED BY

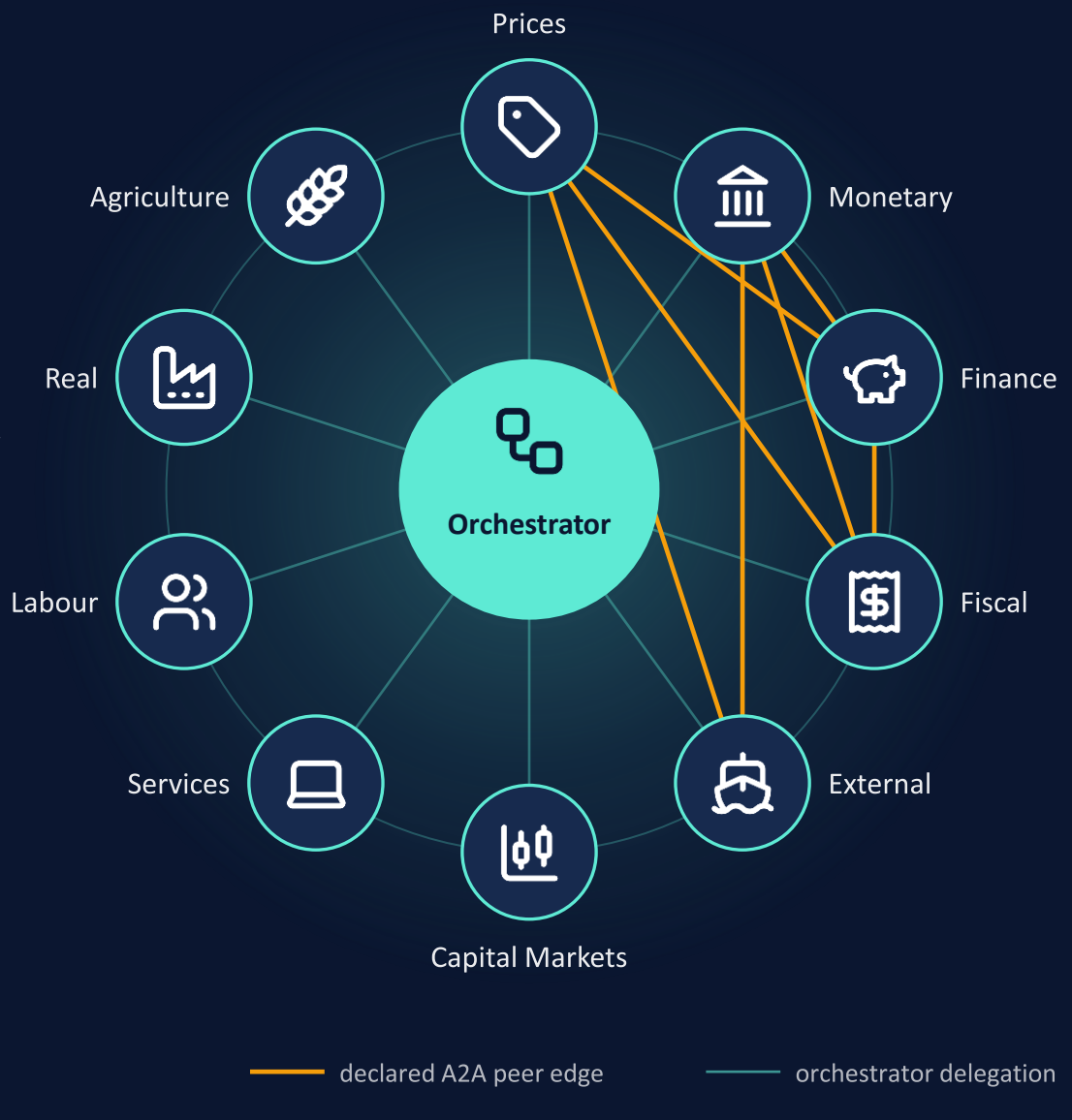
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AGENDA

What this review covers

01

Introduction

Why answers need sources

03 Problem statement

04 Literature survey

05 Objectives

02

System design

How the agents are built

06 Ten sector agents

07 Architecture

08 Technology stack

09 Orchestration

10 A2A trust layer

03

Causal intelligence

How shocks are explained

11 Knowledge graph

12 Scenario engine

13 Causal Lab

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Product and results

What runs and works

15 Product showcase

16 Measured results

17 Key contributions

18 Sample cited answer

05

Outlook

Where it all lands

19 Conclusion

20 Bibliography

One rule runs through every section: **no source, no answer.**

Every figure on these slides names its agent, official table and observation period, and every simulated value says so.

THE PROBLEM

Scattered sources, invented numbers

A TYPICAL ANALYST QUESTION

“How will a $20 rise in Brent crude travel through India’s inflation, interest rates and growth?”

THE ANSWER NEEDS DATA FROM

- **MoSPI** CPI, GDP, IIP
- **RBI DBIE** repo, credit
- **CGA** deficit, capex
- **DGCI&S** trade
- **NSE / BSE** markets
- **IMD** monsoon
- **Agmarknet** mandi prices
- **EPFO** payroll

Different formats, frequencies, base years and release calendars

Fragmented data

MoSPI, RBI, CGA and DGCI&S publish through APIs, spreadsheets and PDF bulletins, each with its own units, base years and release calendar.

Fluent, unsourced numbers

A language model can quote a CPI figure that appears in no official release; because it looks plausible, it is hard to catch.

Conflicting values

When several agents fetch the same indicator on their own, one report can carry two different vintages of it.

Hidden transmission

Shocks move across sectors with lags and signs; describing each series alone cannot explain how a shock travels.

LITERATURE SURVEY

What prior work leaves open

Work	Approach	Gap for Indian macro answers
Hao & Xie (2025)	Several LLMs act as heterogeneous households	Stylised tasks, no observed data
Lin et al. (2026)	LLM agents simulate macro expectation surveys	Simulated survey answers only
Xiao et al. (2025) TradingAgents	Analyst, researcher and trader agent roles	One shared data feed, no owners
Yang et al. (2020)	Macro knowledge graph mined from literature	Links not re-tested on data
Zhu et al. (2025) KG2RAG	Graph-guided expansion of retrieved chunks	The model still writes numbers
Dadopoulos et al. (2025)	Metadata-driven RAG over financial filings	No claim-level verification
Kato (2026) AI Economist Agent	Reports grounded in executed model runs	U.S. only; no Indian sources

Gaps this project targets

Data ownership across agents

One owner per indicator, enforced by the A2A server

Evidence grades for causal edges

Edge audit grades every edge against live official series

Machine-checked numbers

Decimal claim verifier against content-addressed evidence

Coverage of India

MoSPI, RBI DBIE, CGA, DGCI&S behind sector FastMCP servers

OBJECTIVES

Seven objectives, one rule

O1

Sector-wise data access

One FastMCP server and local store per sector, fed by official sources

O2

Coordinated multi-agent reasoning

Typed A2A protocol with discovery, routing, loop guards and tracing

O3

Source-grounded answers

Provenance enforced in the protocol; missing data reported, not estimated

O4

Explicit causal model

Knowledge graph of transmission channels with lag, sign and evidence class

O5

Traceable provenance

Source, table, period, retrieval time and hash on every observation

O6

Research instruments

Edge audit, anomaly radar, trajectory, twin, MPC and claim verification

O7

Usable interface

Cited dashboard, streaming chat, causal lab and agent registry

No source, no answer

Every figure carries agent, table and period; gaps are shown as unavailable

PROPOSED SYSTEM

Ten sector agents, one owner per number

Real
GDP, GVA, IIP, GFCF

MoSPI e-Sankhyiki

Agriculture
Crops, MSP, monsoon, mandi prices

Agmarknet · IMD

Prices
CPI and WPI inflation, food, fuel and core

MoSPI e-Sankhyiki

Monetary
Repo corridor, M3, liquidity, stance

RBI DBIE

Finance
Bank credit, GNPA, WALR and MCLR

RBI DBIE tables

Capital Markets
NIFTY, India VIX, FII/DII flows, G-Sec

yfinance · SEBI

Fiscal
Fiscal deficit, GST, capex, public debt

CGA · Union Budget

External
Forex reserves, trade, BoP, USD/INR

RBI · DGCI&S

Labour
PLFS unemployment, EPFO payroll

MoSPI PLFS · EPFO

Services
Services GVA, PMI, IT exports

MoSPI · PMI releases

A2A for reasoning
agents talk only through typed requests

MCP for data
each agent reads only its own MCP server

No shortcuts
sector packages never import each other

ARCHITECTURE

Reasoning flows apart from data



TECHNOLOGY STACK

Built on an open, typed Python stack

ORCHESTRATION

LangGraph

Four-node StateGraph over a typed shared state

AGENT PROTOCOL

Native A2A layer

Typed Pydantic envelopes, server guards, trace store

DATA TOOLS

FastMCP

One tool server per sector over official sources

API GATEWAY

FastAPI + Uvicorn

REST endpoints and Server-Sent Events chat stream

STORAGE

DuckDB + SQLite

Sector stores, canonical store and claim audit log

GRAPH AND STATISTICS

NetworkX + statsmodels

Path queries, impact queries and Granger tests

LANGUAGE MODELS

Groq gpt-oss-20b / 120b

Fast and reasoning models with offline fallback

FRONTEND

React 18 + TypeScript

Vite, Tailwind, GSAP and Mermaid diagrams

ENGINEERING RULES Python 3.12 with uv · Pydantic v2 everywhere · async I/O with Tenacity retries · parameterised DuckDB SQL · strict TypeScript

ORCHESTRATION

From question to cited report in four steps

1 Route

`decompose`

Scores Agent Cards against the query; an LLM chooses only if no card matches

2 Delegate

`parallel_a2a_execute`

Concurrent sector_analysis requests, plus one bounded follow-up round

3 Enrich

`kg_causal_enrichment`

Shortest causal paths between observed indicators; optional shock simulation

4 Synthesise

`synthesis`

Indicator matrix with provenance hashes, causal diagram and source table

WORKED RUN · OCTOBER 2026

"How does a rise in crude oil prices affect inflation and RBI repo rate policy?"

Shock: Brent +$20 / bbl

Routed to Monetary, Prices and Agriculture

OBSERVATIONS RETURNED

Indicator	Value	Period	Source
CPI headline (YoY)	**5.54%**	Sep 2026	MoSPI
Repo rate	**5.25%**	Feb 2026	RBI DBIE
M3 growth (YoY)	**unavailable**	Jun 2026	RBI DBIE

Every value keeps its period, status and SHA-256 hash

GRAPH PATHS FOUND

CPI → repo · 1 hop, 1 month
Brent → CPI · 2 hops, 3 months
Brent → repo · 3 hops, 4 months

≈ 11 s

end to end with live official data and no LLM key

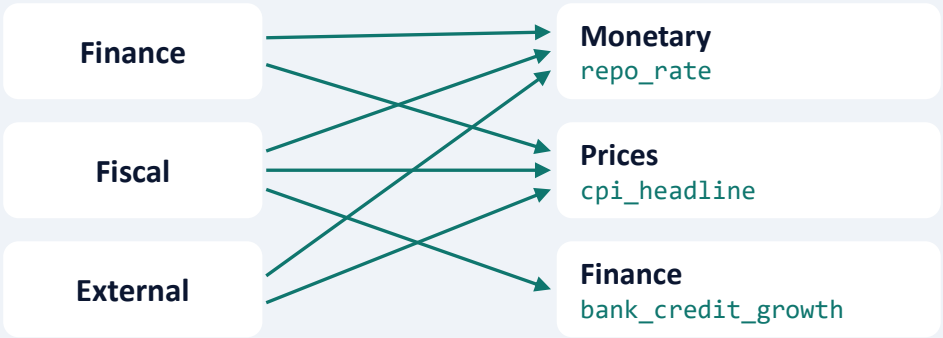
A protocol that refuses unsourced answers

Six checks on every A2A request

1. **Loop guards**
   Depth ≤ 5 · at most 25 requests per conversation · no cycles
2. **Declared dependency**
   Sector-to-sector calls only on the 7 declared edges
3. **Capability**
   The task must be listed in the receiver's Agent Card
4. **Schema**
   Strict Pydantic v2 envelopes; extra fields rejected
5. **Timeout and retry**
   90 s request, 25 s peer; only transient errors are retried
6. **Provenance** No Source, No Answer
   A successful reply with no sources is rejected

7 declared peer edges

consumer → provider · A2A task



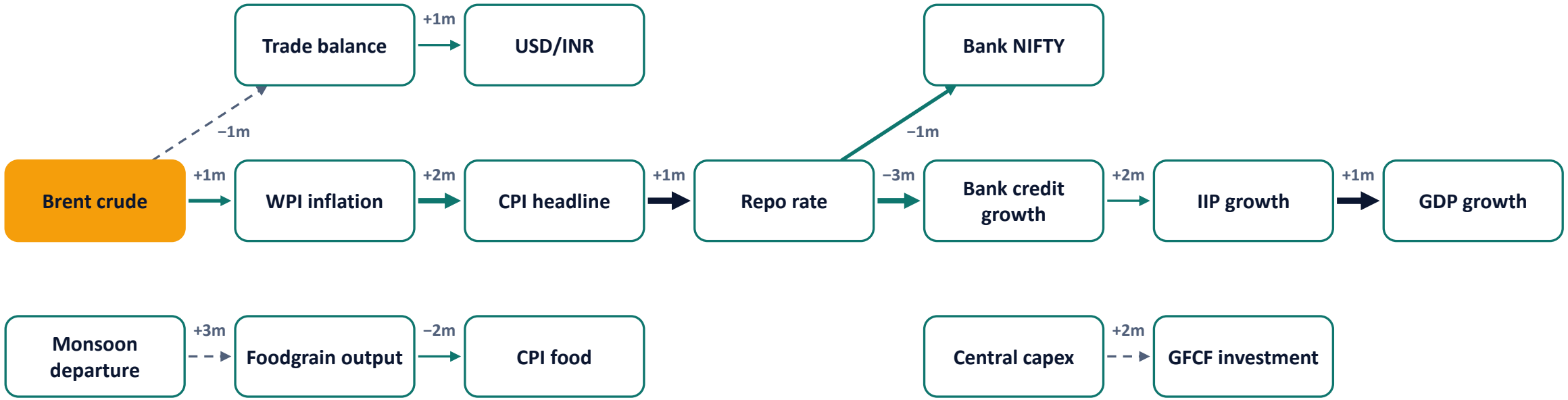
MCP data path for every value

Live official source → Cached snapshot → Structured unavailable

Never an estimate. Each value keeps source URL, dataset, table, period, retrieval time and a record reference.

CAUSAL KNOWLEDGE GRAPH

20 indicators, 12 typed transmission edges



No edges yet

NIFTY 50 · India VIX · Forex reserves · Debt-to-GDP · EPFO additions

EVIDENCE TIER OF EACH EDGE

- Theory
- Statistical association
- Lagged relationship
- Granger predictive
- Structural causal model

EDGE ANATOMY

Each edge: stable rel-id, lag, sign, confidence and mechanism
Checked on load: no self-loops, unknown endpoints, duplicates
NetworkX answers path and impact queries; Neo4j mirror optional

SCENARIO ENGINE

Tracing a $20 oil shock, hop by hop

PROPAGATION RULE

Δ target = shock × Π (sign × 0.55 × confidence) × 0.15

Lags add up along the path, confidence multiplies, and the band is ±25% of Δ.

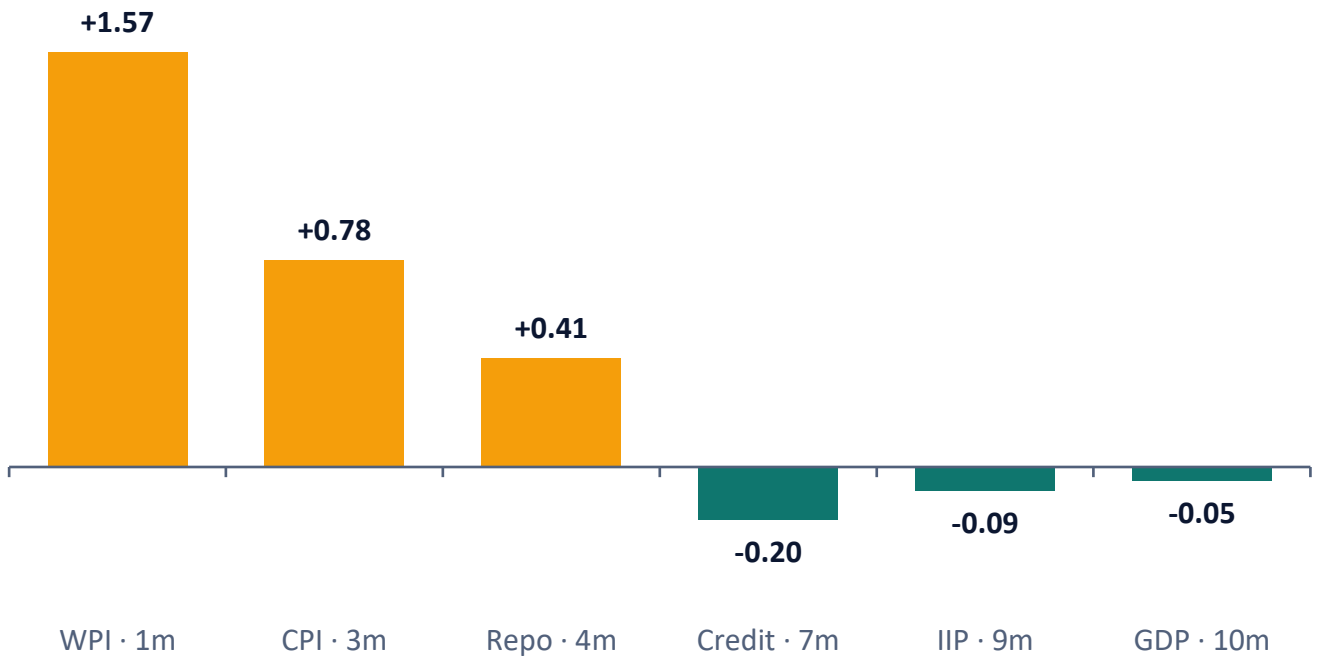
Hop 1: 20 × (0.55 × 0.95) × 0.15 = +1.57
so WPI moves from 2.45 to 4.02

SIMULATED

Transparent by design: every delta traces back to named edges, lags and confidences.

One engine powers Shockwave, Twin and the orchestrator report, so every view tells the same story.

Simulated change along the main chain, percentage points (Brent +$20/bbl)



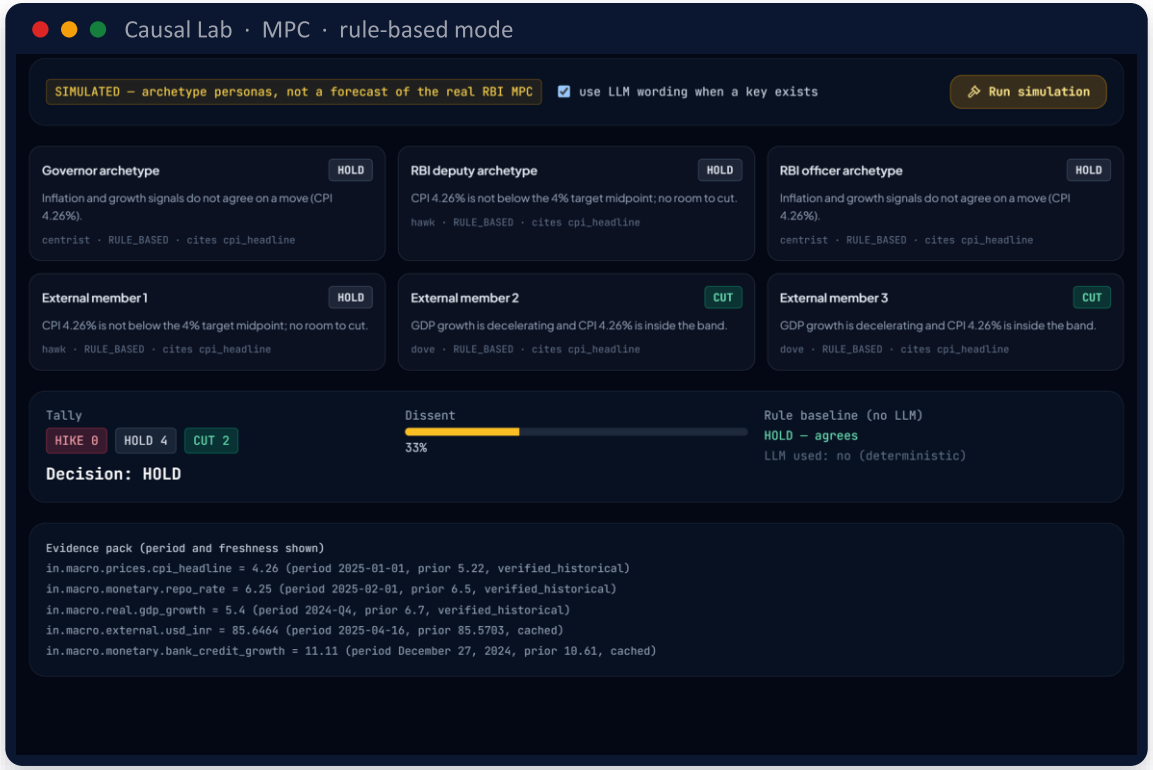
Path confidence 0.95 → 0.85 → 0.81 → 0.71 → 0.59 → 0.57 over six hops

Live baselines live CPI 5.54% (MoSPI, Sep 2026) replaces the default; peak 6.32%

CAUSAL LAB

An interactive lab for every shock



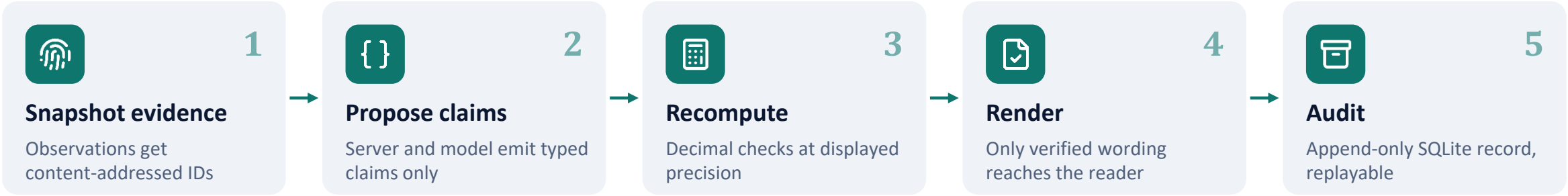


- **Path** shortest route and cumulative lag
- **Shockwave** month-by-month propagation, step or decay
- **Twin** base vs shocked world, weakest hop

- **MPC** six simulated seats vs a rule baseline
- **Radar** z-score anomalies, propagating or isolated
- **Edge audit** Granger tests grade each edge’s evidence

CLAIM VERIFICATION

Numbers checked before they are shown



Status	What it guarantees
`verified`	Numerical claim passed deterministic checks against the snapshot
`source_attributed`	Exact sentence found in a retrieved passage; not a causal proof
`general_mechanism`	Reviewed educational template, not a finding about this period
`rejected`	Bad schema, wrong arithmetic or direction, or an altered quote
`insufficient_evidence`	Missing, conflicting or out-of-scope evidence: claim withheld

DESIGN GUARANTEES

Model prose is never rendered as fact

Unknown claim types cannot bypass the schema

A 6% to 4% move renders as −2 percentage points

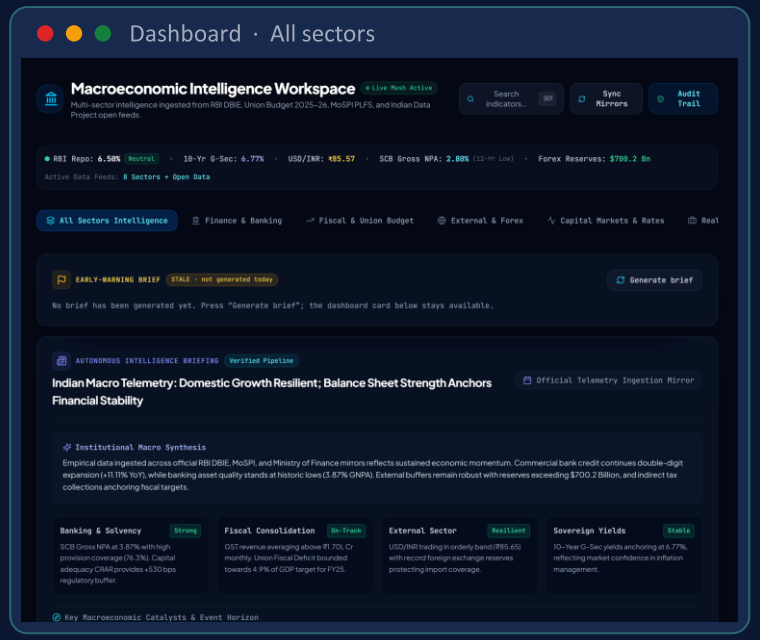
Retrieval time never changes evidence identity

TESTED MODULE

Focused tests cover arithmetic, scope checks, negated quotes and exact audit replay.

PRODUCT SHOWCASE

A working product, not a notebook



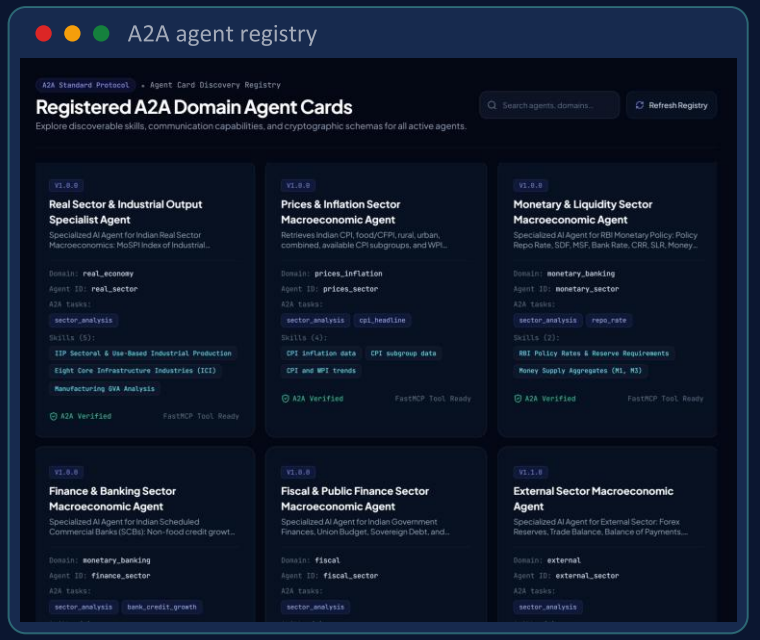
Cross-sector dashboard

Live ticker, early-warning brief and sector tabs; KPIs carry their table and period.



Finance and banking

Credit, asset quality and capital gauges from RBI DBIE tables r539, r330, r531 and r689.



A2A agent registry

Agent Cards with skills, A2A tasks and the declared peer edges between sectors.

ALSO IN THE APP Streaming chat with a live A2A trace · Evidence drawer · Knowledge-graph explorer · Google sign-in · Command palette

RESULTS

What the measurements show

550 / 551

test cases passed across all ten sector suites (455 test functions)

PYTEST · 12 OCT 2026

≈ 11 s

cross-sector answer with live official data and no LLM key

END-TO-END RUN

48.9k

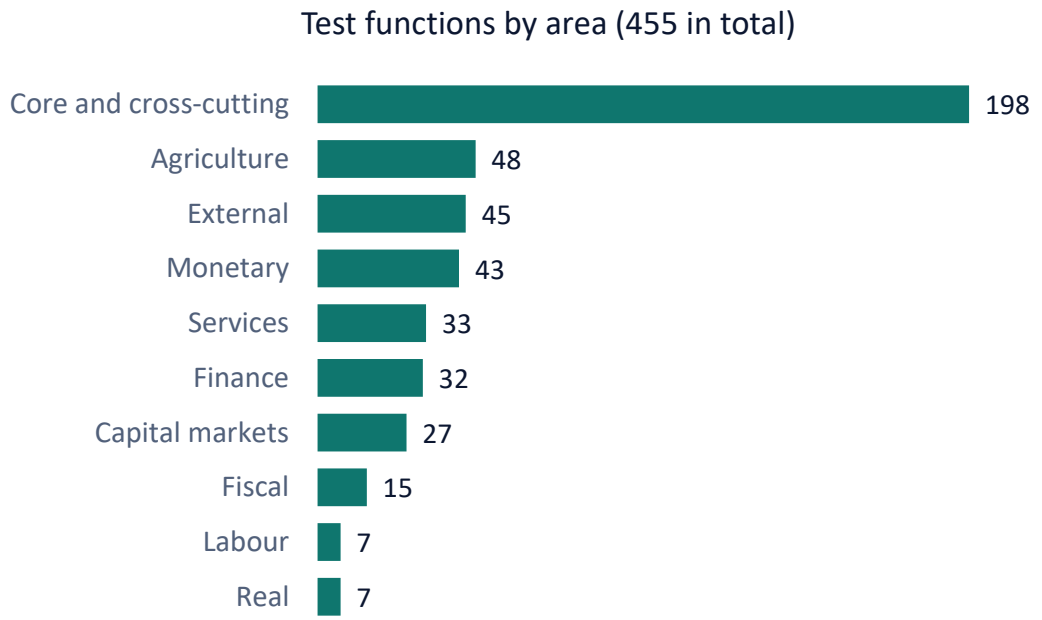
lines of code, excluding tests:
37.6k Python and
11.3k TypeScript

REPOSITORY COUNT

10 / 10

sector agents registered and routable through their A2A Agent Cards

A2A REGISTRY



Anomaly radar

18 series scanned, 8 long enough
Trade balance, Aug 2024: z = −2.65
classified as isolated

Simulated MPC

HOLD by 4 to 2, dissent 0.33
matches the rule-only baseline

KEY CONTRIBUTIONS

What Macrograph AI contributes

1

Ownership-enforced agents

`dependencies.py`

Each indicator has exactly one owner agent; peers obtain it only over the seven declared A2A edges

2

Protocol-level provenance

`A2AServer`

The A2A server rejects any successful reply that carries no source provenance record

3

Evidence-typed causal graph

`knowledge_graph`

Twelve edges on a five-tier evidence taxonomy with stable IDs, lags, signs and an automated audit

4

Deterministic claim verification

`core/verification`

Typed claims recomputed in Decimal arithmetic against content-addressed evidence

5

Official Indian data, end to end

`10 FastMCP servers`

MoSPI, RBI DBIE, CGA and DGCI&S behind ten FastMCP servers, cited down to the table

BUILT TO STAY RELIABLE

Live, cached, unavailable

Every value comes from its source, a tagged snapshot or a clear unavailable status

Works offline

Each stage has a deterministic fallback, so answers arrive even without an LLM key

Fully traceable

Every A2A call carries correlation IDs and joins a conversation trace tree

550 / 551 TESTS PASSING

SAMPLE OUTPUT

What a cited answer looks like

INDICATOR MATRIX

Indicator	Value	Period	Source	Provenance hash
in.macro.prices.cpi_headline	**5.54 %**	Sep 2026	MoSPI (NSO)	fed3aaaa910e5b26
in.macro.monetary.repo_rate	**5.25 %**	Feb 2026	RBI DBIE	37e0e36c25c48aea

SCENARIO TABLE *Brent +$20/bbl, live baselines, labelled simulated*

Target	Baseline	Shocked	Impact	Lag	Band
WPI all commodities	2.45	**4.02**	**+1.57**	1 m	[3.63, 4.41]
CPI headline (live)	5.54	**6.32**	**+0.78**	3 m	[6.12, 6.52]
Repo rate (live)	5.25	**5.66**	**+0.41**	4 m	[5.56, 5.76]
GDP growth	5.40	**5.35**	**−0.05**	10 m	[5.34, 5.36]

SOURCE PROVENANCE

prices_sector NSO, MoSPI · CPI 2026-09 · retrieved 12 Oct 2026, 18:48:49
monetary_sector RBI DBIE · repo rate · retrieved 12 Oct 2026, 18:48:45

EVERY REPORT CONTAINS

1. **Executive synthesis**
   LLM prose or a deterministic summary
2. **Indicator matrix**
   value, period, status, provenance hash
3. **Causal diagram**
   Mermaid graph of the paths found
4. **Transmission paths**
   hops, lags and mechanisms
5. **Scenario table**
   live baselines, impact, lag, band
6. **Source provenance**
   agent, dataset, table, retrieval time

≈ 11 s END TO END

CONCLUSION

Every number has an owner and a source

O1 Data access DELIVERED	O2 Coordination DELIVERED	O3 Grounded answers DELIVERED	O4 Causal model DELIVERED	O5 Provenance DELIVERED	O6 Research tools DELIVERED	O7 Interface DELIVERED

Macrograph AI makes one agent responsible for every number, carries each number with its official source, and explains how a shock travels across the Indian economy, from crude oil to GDP.

NEXT

1. Verify every claim before it is shown
2. Estimate edges from long series
3. Benchmark against five baselines

Thank you

Questions and discussion welcome



github.com/Hariprasaadh/macrograph-ai

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References follow APA style; the full list of 31 sources is in the project report.